

Question3

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.1      ✓ stringr   1.5.2
## ✓ ggplot2    4.0.0      ✓ tibble    3.3.0
## ✓ lubridate  1.9.4      ✓ tidyr     1.3.1
## ✓ purrr      1.1.0
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggthemes)
```

```
transport <- read_csv("https://ourworldindata.org/grapher/co2-emissions-transport.csv")
```

```
## Rows: 6606 Columns: 4
## — Column specification —
## Delimiter: ","
## chr (2): Entity, Code
## dbl (2): Year, Carbon dioxide emissions from transport
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
co2 <- read_csv('data/data_raw/owid_co2.csv')
```

```
## New names:
## Rows: 50407 Columns: 80
## — Column specification —
## ————— Delimiter: "," chr
## (2): country, iso_code dbl (78): ...1, year, population, gdp, cement_co2,
## cement_co2_per_capita, co...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## • `` -> `...1`
```

```
energy <- read_csv('data/data_raw/owid_energy.csv')
```

```
## New names:
## Rows: 23195 Columns: 131
## — Column specification
## _____ Delimiter: "," chr
## (2): country, iso_code dbl (129): ...1, year, population, gdp,
## biofuel_cons_change_pct, biofuel_con...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## • `` -> `...1`
```

```
transport_cr <- transport |>
  rename(
    country      = 1,  # usually "Entity"
    iso_code     = 2,  # usually "Code"
    year         = 3,  # "Year"
    transport_co2 = 4   # emissions from transport, tonnes
  ) |>
  filter(country == "Costa Rica")
```

```
cr_transport <- co2 |>
  filter(country == "Costa Rica", year >= 1990) |>
  select(country, iso_code, year, co2, co2_per_capita, population) |>
  left_join(transport_cr,
    by = c("country", "iso_code", "year")) |>
  mutate(
    # co2 is in *million* tonnes; transport_co2 is in tonnes
    transport_share_total = if_else(!is.na(co2) & co2 > 0,
      transport_co2 / (co2 * 1e6),
      NA_real_),
    transport_co2_pc      = transport_co2 / population,
    non_transport_co2_pc = ((co2 * 1e6) - transport_co2) / population
  )
```

```

cr_trans_plot <- cr_transport |>
  filter(!is.na(transport_share_total),
         year >= 1990)

p3 <- ggplot(cr_trans_plot,
             aes(x = year)) +
  geom_line(
    aes(y = transport_share_total * 100),
    colour = "#D55E00",
    linewidth = 1.1
  ) +
  geom_hline(yintercept = 0, linetype = "dotted", colour = "grey70") +
  labs(
    title = "Transport's share of Costa Rica's CO\u2082 emissions",
    x = "Year",
    y = "Transport share of total CO\u2082 (%)"
  ) +
  theme_clean(base_size = 12) +
  theme(
    panel.grid.minor = element_blank()
  )

```

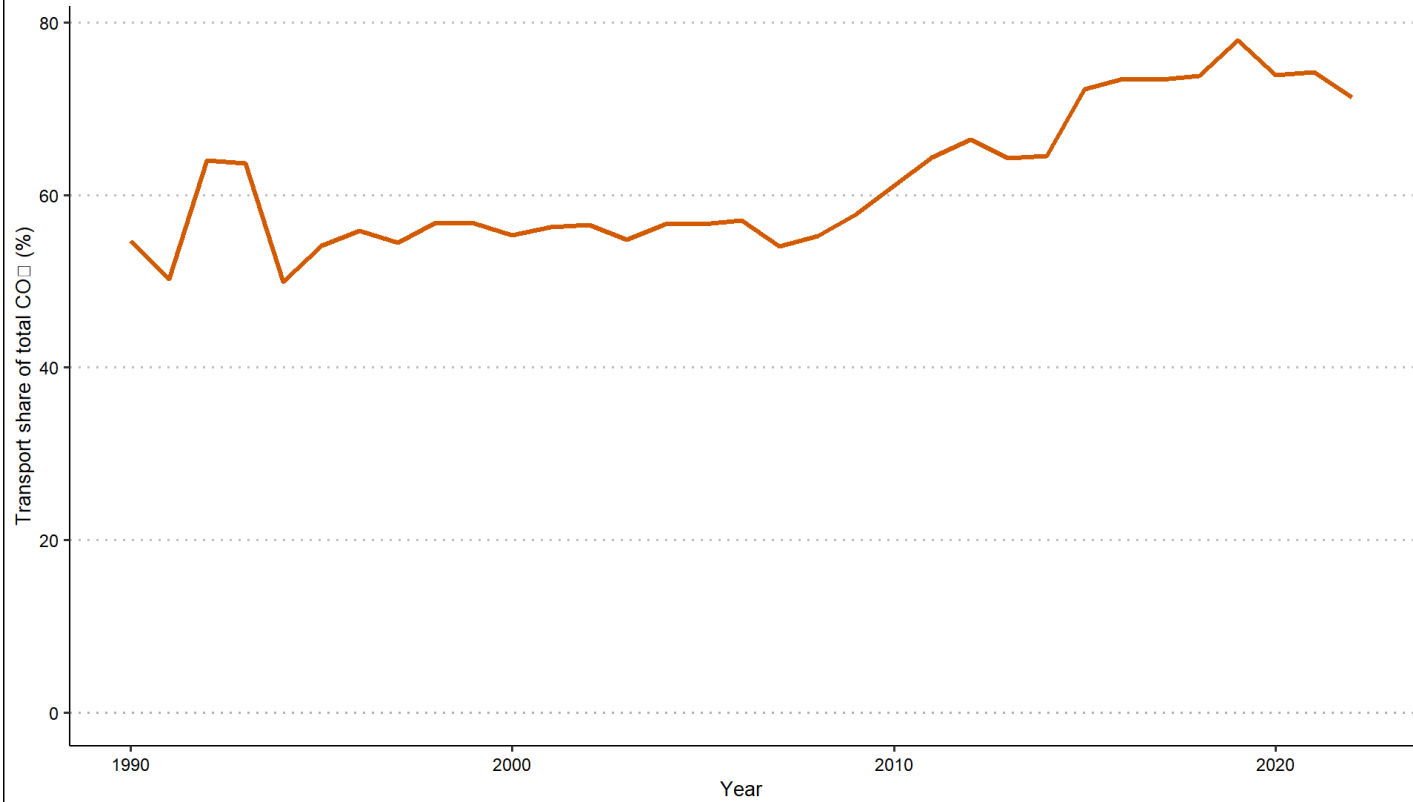
```

## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## i The deprecated feature was likely used in the ggthemes package.
##   Please report the issue at <https://github.com/jrnold/ggthemes/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

p3

Transport's share of Costa Rica's CO₂ emissions



```

cr_pc_long <- cr_transport |>
  filter(!is.na(transport_co2_pc),
         year >= 1990) |>
  select(year, transport_co2_pc, non_transport_co2_pc) |>
  pivot_longer(
    cols      = c(transport_co2_pc, non_transport_co2_pc),
    names_to  = "sector",
    values_to = "co2_pc"
  ) |>
  mutate(
    sector = recode(
      sector,
      transport_co2_pc      = "Transport",
      non_transport_co2_pc = "All other sectors"
    )
  )

p4 <- ggplot(cr_pc_long,
             aes(x = year, y = co2_pc, colour = sector)) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = c(
    "Transport"      = "#D55E00",
    "All other sectors" = "#0072B2"
  )) +
  labs(
    title = "CO\u2082 per capita from transport vs other sectors",
    x = "Year",
    y = "Tonnes CO\u2082 per person",
    colour = NULL
  ) +
  theme_clean(base_size = 12) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
p4

```

CO₂ per capita from transport vs other sectors

